A Unified Study of Influence Maximization in Social Networks: HIM, DeepSN, and GBIM

Course Project Milestone Report
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Abstract

This milestone defines a unified course project on social-network information diffusion and influence maximization. Three members prepare complementary methods: HIM learns hyperbolic user representations from a graph and propagation histories; DeepSN uses a sheaf-neural surrogate for diffusion-aware seed selection; and GBIM applies Bayesian optimization to multiplex user–item influence. HIM and DeepSN now both have verified team environments and end-to-end experiments. On Cora-ML, one fixed-seed HIM WIC+ASW run produced 37.6195% independently evaluated coverage at its saved final epoch. DeepSN reproduced the authors' Jazz SIS protocol at 86.4495% coverage and completed a Ciao three-seed WIC study aligned with HIM's graph, probabilities, budget, and evaluation stream (mean coverage 8.132%). A paired HIM ASW/MD comparison found a small conditional ASW advantage; neither Ciao three-seed interval supports a between-method ranking. We select the partially released Ciao review/trust network as the primary new-data target because it combines user trust relations with user–item interactions, connecting the user-level methods to GBIM's native multiplex setting. Direct three-way ranking will be reported only if the GBIM adaptation passes an explicit objective-equivalence check.

1 Problem and Project Scope

We select the project theme of social-network transmission analysis, with influence maximization as the central task. Given a graph $G = (V, E)$, a diffusion process $\mathcal{M}$, and a budget k, the goal is to choose an initial seed set:

$$S^* \in \arg\max_{S \subseteq V,\ |S| \le k} \sigma_{\mathcal{M}}(S), \qquad \sigma_{\mathcal{M}}(S) = \mathbb{E}\left[|A_{\mathcal{M}}(S)|\right], \tag{1}$$

where $A_{\mathcal{M}}(S)$ is the set eventually activated from S. The practical difficulty is that the diffusion rules, edge probabilities, observed histories, and even the meaning of a "seed" differ across methods. Selecting high-degree users alone does not solve the set problem: two individually strong users may reach much of the same audience.

Our overall question is: *How do graph-based influence methods use structural, propagation, and user–item evidence to select effective interventions under a limited budget?* The primary dataset is the Ciao review/trust network. The methods are linked through one data pipeline and a staged experiment: first compare user-level seed selection under one fixed propagation evaluator; then test whether item-aware user–item actions improve reach on the same users and trust graph. This preserves a common question without pretending the original objectives are interchangeable. DeepSN's native authors' protocol uses SIS on Jazz, whereas the shared Ciao contract uses weighted independent cascade (WIC). Those two numbers answer different questions and must never be pooled. The intended contribution is a reproducible comparison pipeline, a careful explanation of information assumptions, and a small, interpretable empirical study rather than a claim of universal superiority.

2 Related Methods and Their Connection

Classical influence maximization maximizes expected spread under an explicit diffusion oracle. Under standard assumptions, the spread objective is monotone and submodular, allowing greedy selection to obtain a $1 - 1/e$ approximation with an exact marginal-gain oracle [1]. Contact-process (SIS) dynamics allow recovered nodes to become susceptible again, so a reported "infected share" need not correspond to an absorbing activated set [2]. The methods assigned in this project move beyond a single greedy oracle:

- **HIM** learns one user embedding from social edges and observed propagation activations. Its Lorentzian distance to the origin estimates individual influence strength; its adaptive sliding window (ASW) adjusts scores of nearby candidate neighbors to reduce likely overlap [3].
- **DeepSN** treats influence as a learned map from a seed indicator to an activation pattern. Restriction maps of a cellular sheaf allow neighbor-to-neighbor influence to be heterogeneous rather than a shared scalar weight [5, 6, 4]. Seed assignment then uses community-proportional budget allocation rather than a single radial score.

- **GBIM** uses graph Bayesian optimization for multiplex influence maximization. Its native decision is over user–information pairs on user and item-related graphs, not simply a set of users [7].

DeepIM offers a related seed-set representation-learning approach, but is not one of the three assigned implementations in this project [11]. The methods can be presented as complementary views of a single pipeline: observed graph and interaction data; representation or surrogate learning; seed/action selection; and independent cascade evaluation. HIM emphasizes geometric influence ranking plus overlap correction, DeepSN emphasizes a richer neural model of propagation, and GBIM emphasizes sample-efficient optimization in a multiplex action space. They should not be combined into an ensemble merely because they share the phrase "influence maximization."

3 Data Collection and Initial Findings

3.1 Existing evidence

The verified HIM reproduction uses the author-provided Cora-ML graph with 2,810 nodes and 7,981 undirected edges. Its WIC training cache contains 30 propagation instances. The saved run uses WIC, ASW, seed ratio 10% ($k = 281$), training seed 42, and stops at epoch 54. The saved final-epoch seed set was independently evaluated for 1,000 simulations with evaluation seed 1042: mean activated users $1,057.107$, standard deviation 32.1214, and coverage 37.6195%. Reloading the model reproduced the exact seed set and evaluation summary. A distinct maximum validation coverage of 38.15% occurred at epoch 43; it must not be reported as the independent score of the saved epoch-54 model.

The reproduction must also name the actual simulator configuration. The paper specifies WIC probabilities in $[0, 0.2]$, but the bundled Cora-ML probability file spans approximately $[0.1, 0.2]$. The paper describes WLT parameters as fixed across propagation realizations, whereas the repository resamples node thresholds during each WLT simulation. We therefore treat the verified run as a reproduction of the recorded code/data setup, not as an exact match to every manuscript setting.

An exploratory temporal Email-Eu-core Department 3 sample has also been downloaded from SNAP [10]. The recorded source is https://snap.stanford.edu/data/email-Eu-core-temporal.html. The current pilot covers 89 users and 3,514 events in a 180-day window; retaining directed edges observed at least twice gives 464 edges, 15 isolates, and 16 weakly connected components. This is a useful pipeline smoke test, but it is small and fragmented, has no message text or sentiment labels, and its email events are not validated diffusion cascades. It cannot by itself support claims about real opinion adoption.

3.2 DeepSN native Jazz collection

DeepSN's author protocol is tied to the Jazz collaboration network shipped with the public repository (commit `bbc7780`), not to Ciao. The binary cache `jazz_mean_SIS200.SG` is downloaded from that commit and kept local (SHA-256 prefix `8e7f1eed`): 198 nodes and 100 seed–activation training pairs under SIS. It is not committed to Git. On an Apple Silicon Mac Air, Python 3.11.17 and PyTorch 2.5.1 CPU, one native run with 20% budget ($k = 40$), training seed 42, 50 estimation plus 200 scoring epochs, and 100 SIS evaluations produced mean coverage 86.4495% (standard deviation 2.173 percentage points; mean activated 171.17). That figure only describes a dense 198-node contact process. It is not a Ciao result and is not comparable to HIM's Cora-ML WIC score.

3.3 Selected primary dataset and collection procedure

Ciao is selected as the primary new dataset because it provides a trust graph between users and user–item review/rating relations. We retrieved the baseline MAT archive from the dataset authors' page [9]; the page says the collections were crawled in May 2011 and that the published release is only a partial version. The archive contains 284,086 ratings and 111,781 directed trust arcs. The trust graph covers 7,317 users; its undirected projection has 85,205 edges, seven connected components, and a largest component of 7,305 users. Rating records cover 7,375 users, of whom 58 have no trust-network membership; the shared graph-aligned rating table therefore retains 283,320 source rows. The Ciao page and accompanying README do not state an explicit data license or redistribution permission. We keep raw MAT files locally, do not commit or redistribute them, cite the source, and will confirm course-use expectations before any public release. Full provenance, hashes, schemas, and counts are recorded in `data/ciao_source_manifest.json`.

The baseline MAT archive has no rating timestamps, and trust links do not have establishment dates. A separately packaged timestamped text release has 57,544 trust rows and 36,065 timestamped rating rows, substantially fewer than the baseline MAT archive. We have not established that it is a record-aligned timestamp extension, so it is excluded rather than merged. A shared preprocessing script now creates sorted contiguous user/item maps, preserves directed trust edges, exports a deduplicated undirected user graph, and retains original in-scope rating rows (including duplicate pairs and zero ratings). GBIM's existing adapter instead keeps the first rating per user–item pair, removes zero ratings, filters users outside the trust graph, and symmetrizes trust edges; that yields 281,865 ratings over 7,317 users and 104,957 items. Both views are documented so method-native behavior is not silently conflated with the shared data export.

Table 1: Method roles, inputs, and comparison boundary.

Method	Native decision	Main evidence	Fair-comparison requirement
HIM	k user nodes	Social graph and propagation instances	Fix graph, cascade bank, budget, and independent evaluator; compare ASW with HIM-MD.
DeepSN	k user nodes / learned seed set	Graph and generated or observed diffusion examples	Use the real CLI; set sheaf `graph_size` to $\|V\|$; train on a cascade bank separate from the evaluator; report SIS/Jazz and WIC/Ciao as distinct protocols.
GBIM	User–item seed pairs	Social and user–item/item relations	First reproduce its native multiplex objective; adapt to one shared user-only task only after validating the changed objective and code.

The trust graph gives HIM and DeepSN a common user-level network; the interaction layer gives GBIM the multiplex structure its native method expects. Ratings express product feedback, not evidence that one user caused another user's adoption. Since the dataset contains no observed diffusion cascades, user-level influence experiments use a fixed, documented synthetic IC/WIC generator on this real topology, with training and evaluation cascade banks kept separate. The shared WIC rule is

$$p(u, v) = \frac{1}{\max(\deg u, \deg v)}, \tag{2}$$

identified in the shared protocol file as `ciao_symmetric_weighted_cascade_v1`. DeepSN does not reuse HIM's training cache file: for each training seed $s \in \{42, 43, 44\}$ it samples 30 independent WIC histories, writes a method-local `.SG` cache, trains, then evaluates the saved seed set with 1,000 WIC rounds and evaluation seed base 1042. HIM's first smoke test used the initial 5% budget (366 users, nearest-integer rounding); DeepSN uses the same $k = 366$. Add 1% and 10% only after all method adapters pass the same input and output checks. GBIM's item-aware action remains a separate native-task extension unless it can be shown to share the user-only decision space and objective.

3.4 DeepSN Ciao summary statistics

Table 2 records the independent WIC evaluations. Across three training seeds the mean activated set is 595.05 users (sample sd 20.30), or 8.132% coverage; the 95% t interval with two degrees of freedom is $[544.63, 645.47]$. Seed 44 (571.62) is well below seeds 42 and 43 ($\approx$ 606.5–607.1), so the surrogate is sensitive to the 30-cascade draw. HIM-ASW on the same graph, probabilities, k, and evaluation stream has three-seed mean 735.03 (10.05%). The trainers differ (DeepSN: 50+200 epochs, no HIM-style 100-round online validation), so the gap is not an algorithm ranking. A native DeepSN IC evaluator (p = 1/in-degree) on the same Ciao seeds reports about 26% coverage; switching the evaluator changes the task.

Table 2: DeepSN independent WIC on Ciao ($k = 366$, 1,000 rounds, eval seed 1042).

Train seed	Mean $\|A(S)\|$	Coverage %	Train seconds
42	606.47	8.288	188.6
43	607.06	8.297	196.5
44	571.62	7.812	204.3
Mean	595.05	8.132	—

4 Mathematical Background and Algorithms

For independent-cascade (IC), each newly active node gets one attempt to activate each inactive neighbor, with edge probability p_{uv}. For linear-threshold (LT), node v activates once the total incoming active-neighbor weight reaches its threshold θ_v. SIS instead allows nodes to return to a susceptible state and be reinfected, so the reported quantity is a time-averaged infected share rather than a terminal set [2]. These models permit controlled evaluation; they do not prove that real users follow either mechanism. Expected spread is expensive to compute exactly, which motivates simulation, approximations, or learned surrogates.

HIM uses a Lorentz hyperbolic space, where negative curvature can represent hierarchical structure compactly, following a family of hyperbolic embedding methods [12]. It learns user embeddings by fitting social edges and activation edges with relation-specific rotations and negative sampling. A proactive activation term moves users observed to activate others closer to the Lorentz origin. Its strength proxy is

$$\mathrm{LDO}_u = d^2_{\mathcal{L}}(\mathbf{x}_u, \mathbf{o}_{\mathcal{L}}), \tag{3}$$

where a smaller LDO implies higher estimated influence. ASW begins with ascending LDO scores and a candidate window of size $w \approx \beta k$. After selecting seed c, it increases scores of candidate neighbors according to their hyperbolic proximity to c; larger updated scores make them less likely to be selected next. This is a local approximation to audience overlap, not exact marginal spread, so no

greedy approximation guarantee follows automatically.

DeepSN replaces repeated marginal-gain oracles with a two-stage learned procedure [4]. A cellular sheaf assigns a stalk $\mathbb{R}^d$ to each node and a linear restriction map $F_{u\to v}$ to each oriented edge; the sheaf Laplacian L_F induces a diffusion process that Neural Sheaf Diffusion parameterizes from node features [5, 6]. Stage 1 (estimation) takes sparse adjacency and seed–activation pairs. Node features are local structural one-hots (two-hop degree bins) mapped to a fixed width. A sheaf block, together with learnable sigmoid edge weights on the support of G, outputs a surrogate activation $\hat{a} = f_\theta(s)$. The estimation loss is the discrepancy between $\hat{a}$ and the training activation. Stage 2 freezes f_θ, builds a graph from the learned edge weights, partitions it with Louvain, and trains a shallow scoring network. Budget k is split in proportion to community size, with remainder filled by globally high-scoring nodes. The assignment is a hard 0/1 indicator, so gradients are weak; the scoring network mainly ranks nodes into positions the proportional rule can hit. The returned set is then evaluated by Monte Carlo under a named simulator. The course implementation reads the real command line, stops sheaf argument parsing from stealing `-d`, and sets sheaf `graph_size` to the current $|V|$ instead of the Jazz constant 198. Author training and selection loops are otherwise unchanged.

The experimental design will separate *training information* from *evaluation information*. Classical Greedy/CELF can be included as known-parameter reference methods, but must be marked as receiving the true simulator parameters if the learned methods only see generated histories. Degree and random selection provide inexpensive controls. The primary metric is mean spread ratio $100\,\mathbb{E}[|A(S)|]/|V|$, with confidence intervals across independent training seeds; we will separately record data-preparation, training, seed-selection, and evaluation time. The number of propagation simulations, graph, seed budget, and evaluation random seeds will be fixed across methods.

5 Current Progress, Difficulties, and Work Plan

The main risks are: (i) the three methods do not optimize identical objects; (ii) reproducing results depends on diffusion parameters and whether histories are shared; (iii) data transformations can silently change graph direction, node identity, or the seed budget; (iv) training randomness and finite Monte Carlo evaluation add uncertainty; (v) public ratings/trust data may not encode actual information transmission; and (vi) the Ciao page does not state an explicit license. DeepSN adds three implementation risks that already affected this milestone: a default SIS simulator that the original CLI did not actually switch; a sheaf module that parsed `-d` as stalk dimension and collided with the dataset flag; and a hard-coded Jazz `graph_size`=198 that would have made Ciao mathematically wrong. Torch 2.14 also crashed against the published PyG `torch-sparse` wheels on Apple Silicon, which is why the lock is Torch 2.5.1. We will report these as design limitations instead of hiding them behind a single leaderboard.

Table 3: Team status at this milestone.

Workstream	Evidence and remaining work
HIM	Cora-ML independent evaluation and same-checkpoint ASW/MD ablation verified; Ciao adapter, fixed synthetic WIC bank, and three 30-instance training seeds plus 1,000-round held-out ASW/MD evaluations complete. Across-seed paired interval includes zero and is imprecise; add degree/random controls before comparison.
DeepSN	Mac Air CPU environment verified (Python 3.11.17, Torch 2.5.1). Jazz SIS native protocol complete (86.4495%). Ciao uses the shared undirected trust graph and WIC rule; three training seeds, 30 cascades each, independent 1,000-round WIC (mean 8.132%). Next: degree/random controls and more training seeds to shrink the t interval.
GBIM	Author code and LastFM/Delicious data are present; environment and training are not yet verified. Reproduce native Multi-IC first; its user–item objective needs explicit treatment in any shared ranking.
Team	Ciao files and schema are audited and the shared preprocessing entry point is in place; verify data-use expectations, add missing controls, freeze the shared evaluator, then complete GBIM's native and (if valid) user-only adapters.

Proposed milestones. Confirm with the instructor that local educational use of the Ciao release is acceptable; preserve the existing shared export and validate row counts and mappings on each member's machine. HIM's Ciao adapter and three-seed synthetic WIC experiment are complete; DeepSN's matched-protocol three-seed independent evaluation is also complete. Next add Degree/Random controls on the same WIC evaluator for both user-level methods, and expand DeepSN from three to five training seeds so the coverage interval is not dominated by seed 44. GBIM should reproduce its native Ciao objective. Test whether GBIM can be constrained to the same user-only action space without changing the task; if not, retain its native objective as the multiplex extension and report it separately. Then run the shared user-level benchmark with a common training bank where applicable and at least three independent training seeds; extend to five seeds and additional budgets after the pipeline is stable. Individual reviews are in progress (HIM English; DeepSN Chinese by Li Zixi). The exact final-project date remains to be confirmed with the instructor.

6 Conclusion

The project is feasible if it is framed around one selected Ciao data source and a staged comparison of complemen-

tary influence-maximization paradigms. HIM has a verified Cora-ML starting result and a small same-checkpoint selection ablation; on Ciao, three 30-instance HIM training runs and their independent paired evaluations are complete, but they do not establish a selector advantage or cross-method ranking. DeepSN has a verified native Jazz SIS result and a Ciao three-seed WIC evaluation that shares HIM's graph, probability identifier, $k = 366$, and evaluation seed 1042 (mean coverage 8.132% versus HIM-ASW 10.05%). That juxtaposition is a protocol match, not a ranking: trainers, online validation, and selectors differ, and DeepSN's across-seed interval is wide. GBIM has not yet completed its Ciao checks. The existing 89-user Email-Eu-core pilot is better treated as a parser/smoke test. The next high-value actions are to confirm data-use expectations, add degree/random controls, complete native GBIM checks, and freeze the common user-level evaluation contract. The GBIM multiplex result then extends the same study to item-aware interventions without obscuring objective differences.

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